Music Genre Classification via Dimensionality Reduction, Mutual Information Feature Selection, and Causal Analysis

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Abstract

This project investigates dimensionality reduction and classification strategies using the prediction$_m$usic$_g$enredataset*. We begin with an exploratory analysis of the dataset's statistical properties and feature distributions. Subsequently, we evaluate the efficacy of Principal Component Analysis (PCA), Kernel PCA (KPCA), and a suite of Information Theoretic feature selection methods—including MIM, mRMR, and JMI—in optimizing feature spaces for music genre prediction. These techniques are benchmarked using k-Nearest Neighbor and Random Forest classifiers. Finally, we examine the causal relationships between variables and assess the model's robustness under interventional scenarios.*

1 Data and Preprocessing

We inspected the dataset's properties in the Classical and Rap genres, noticing it contains "?" instances in the tempo column. Also, some durations are invalid numbers, for example negative 1.. The removal of missing values resulted would result in a 19 per cent shrink in the dataset, so we applied median input ation on these values.

We remove the obtained-data column, as well as instance-id. The track-name and artist-name could become a feature for language modelling, as we expect some structure to be present distinguishing each genre, but it would probably result in non interpretable classification. We encode the categorical values like key and mode.

2 Features

Before spliting the dataset with the usual seed 42, we noticed class inbalance was minimal (0.15 per cent relative difference between classical and rap and a maximum of 1.5 per cent between Alternative and Rock) and so determined stratification wouldn't be necessary, as we don't expect a bias to develop inside the classifier.

Initially we look closer at the distribution of numeric features, namely instrumentalness, that between these two classes will probably be the least informative, as it is squashed to zero on all datapoints, as seen in Figure 1. We point out that features like energy, danceability and in particular acousticness clearly differentiate the two genres, conjecturing that they are designed to capture musical caractheristics that may correlate strongly with genre. That being the case, a simple decision tree algorithm may already capture a meaningful amount of the separation, although ensembles are likely to perform better, as the cost of the latter scales with the number of genres due to the necessity for informed heuristics. Adding to that, features like duration and liveness are very concentrated throughout the different genres, making them less useful for these kinds of approaches. We could construct a composite score reflecting the degree to which a track exhibits characteristics associated with rap, that could become a rapiness feature, though in practice a supervised model shoul learn such interactions more reliably.

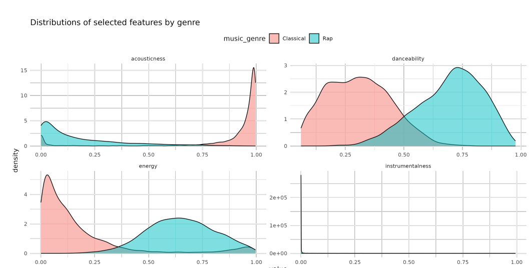

Figure 1: Distributions of acousticness, danceability, energy and instrumentalness over the Rap and Classical genres

We notice that some features are roughly unimodal

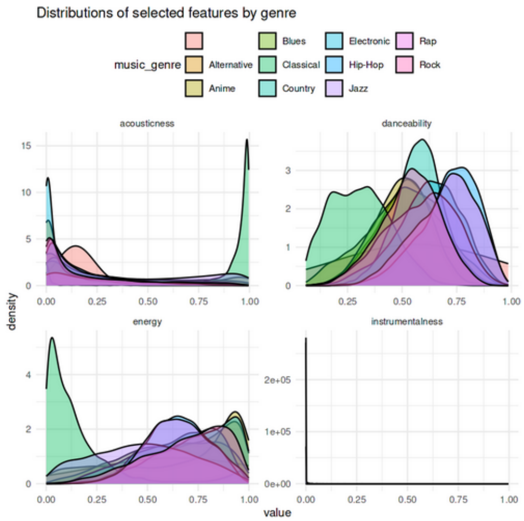

Figure 2: Distributions of acousticness, danceability, energy and instrumentalness over all genres

and bell-shaped, namely danceability and popularity, whereas many others divide the data into two clusters of very high values and very low values, like acoustisness, taking discrete binary values essentially, being highly skewed. Besides these some are bounded, or concentrated near zero, like duration and instrumentalness, respectively.

THe audio features are not orthogonal by construction (even before a dataset is assigned). For example, `energy` takes into account loudness and tempo [6], which directly explains the oriented edge between `energy` and `loudness` recovered by the PC algorithm (Section 5). Some features are presumably calculated through an explicit formula (popularity), while others (instrumentalnes and acousticness) are confidence measures output by a classifier [6]. Still, richer feature documentation could improve the interpretability of any downstream classifier trained on this data.

3 Methods and Results

3.1 Principal Component Analysis - PCA

We first applied Principal Component Analysis (PCA) to reduce the dimensionality of the dataset. This method identifies an orthogonal basis where the axes describe the variance in descending order [8]. As shown in Figure 3, the PCA algorithm effectively reduced the problem dimensionality from 11 to 6 variables while retaining approximately 8% of the total variance. This reduction by 5 dimensions represents a significant compression with minimal information loss. The specific linear combinations for the first two principal components, which capture relationships within the data, are illustrated in Figure 4.

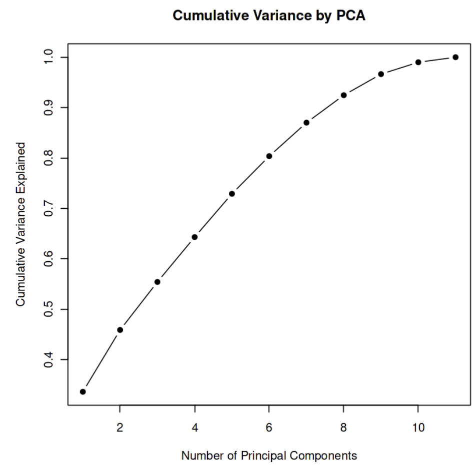

Figure 3: Distributions of acousticness, danceability, energy and instrumentalness over all genres

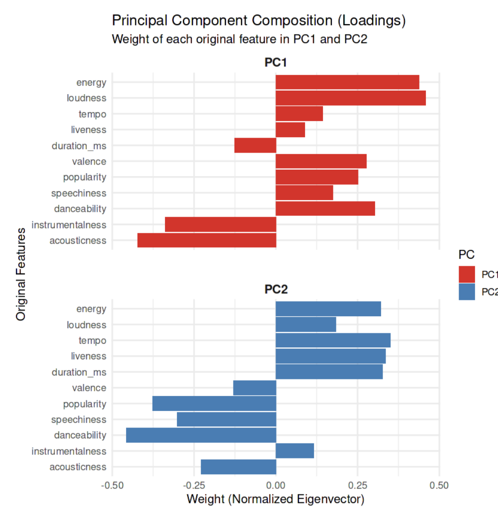

Figure 4: Distributions of acousticness, danceability, energy and instrumentalness over all genres

While the PCA method identifies the axes of maximum variance, the deeper value of Figure 4 lies in uncovering the latent relationships between features. Since PC1 captures the primary variance, its loadings reveal the dataset's fundamental structure: a clear synergy between energy and loudness, which stand in direct inverse relation to acousticness and instrumentalness. Furthermore, the negligible weights for duration and liveness suggest these variables are largely independent of this primary "intensity" axis. PC2 then refines the analysis by capturing "residual" variance, describing secondary characteristics—such as the contrast between technical complexity and commercial appeal—that remain after the primary relations are accounted for.

3.2 KPCA

To further refine our analysis, we implemented Kernel Principal Component Analysis (KPCA) using three distinct kernels: Radial Basis Function (RBF), Polynomial, and Laplacian [8].

Since these kernels are sensitive to hyperparameter selection, we employed an optimization function to identify parameters that maximized the variance retained within the first two principal components. Due to computational complexity and resource constraints, this optimization was performed on a representative 20% random subsample of the training data [9]. The resulting optimal parameters are summarized in Table 1, while the full range of tested parameters is documented in Appendix A.

Table 1: Hyperparameter optimization and cumulative variance for Kernel-based methods.

Kernel Type	p1	p2	2axis_info (Cum. Var.)
RBF	0.001	NA	0.4521
Polynomial	4	0	0.9998
Laplacian	0.001	NA	0.2876

o apply the Kernels to the full training set, we addressed the computational challenge of processing 35000×35000 matrices by again employing the Nyström Method [9], utilizing a 30% random data subsample to compute the first four PCs. The resulting projections and cumulative variances are presented in Figure 5 and Table 10. Notably, the Polynomial kernel concentrated nearly all variance into a single axis. In contrast, the RBF kernel yielded results comparable to standard PCA, while the Laplacian kernel performed the poorest, capturing only approximately 40% of the total information within the total components.

3.3 Information theoretic aware Feature Selection

Eight sequential forward selection methods based on mutual information were implemented and applied to the 11 scaled numeric training features: MIM, MIFS, mRMR, maxMIFS, CIFE (implemented here as its normalised variant DISR), CMIM, JMI, and DMIM [3, 4, ?]. All methods share the same skeleton: at each step the candidate feature X_i maximising a method-specific scoring function $J(X_i)$ is appended

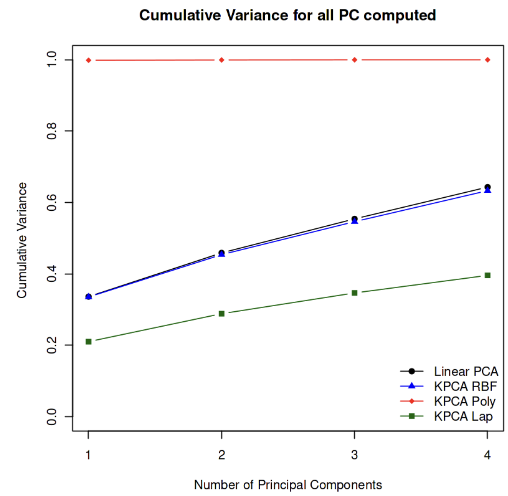

Figure 5: Cumulative variance for all PCA and KPCA methods until 4 components

to the selected set S. The functions differ in how they trade off relevance $I(X_i; C)$ against redundancy with already-selected features, and whether they incorporate class-conditional complementarity $I(X_i; X_s \mid C)$.

MI was estimated via equal-frequency histogram discretisation with $k = \lceil \sqrt{N} \rceil$ bins [?] for the univariate case, with analogous bin counts for joint and conditional distributions.

The number of features to retain (k_{final}) was determined by inspecting figure 3.

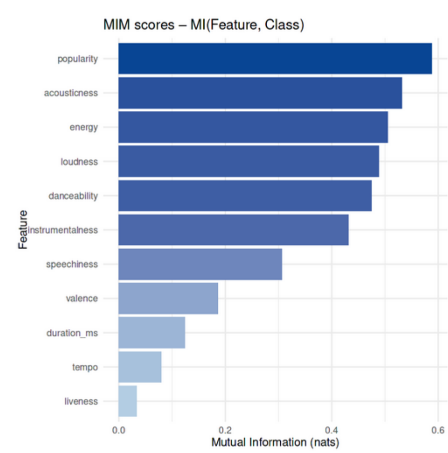

Figure 6: MIM relevance scores for all 11 numeric features. The elbow after position 6 justifies retaining $k_{\text{final}} = 6$ features across all methods.

Table 1 reports the full feature rankings produced by all eight methods. There is broad agreement on the top positions: `popularity` and `loudness` are consistently placed in the top two or three, reflecting their strong genre-proxy role `Instrumentalness`, danceability, and speechiness appear in the top six for most methods, capturing the known acoustic signatures of Classical (near-binary high instrumentalness), Rap and Hip-Hop (high speechiness and danceability), and Electronic music. The bottom three features (duration_ms, tempo, liveness) are never selected, confirming their low marginal relevance. The key divergences are in positions 6–8: MIFS uniquely promotes danceability to rank 2 because its unscaled penalty pushes the method away from loudness-correlated features early in the selection process; CMIM uniquely selects valence, the only method to do so, reflecting its complementarity-aware scoring.

Table 2: Feature rankings for five representative methods.

Rk	MIM	MIFS	mRMR	CMIM	DMIM
1	pop.	pop.	pop.	pop.	pop.
2	loud.	dance.	loud.	loud.	loud.
3	acous.	loud.	instr.	instr.	instr.
4	instr.	instr.	dance.	dance.	dance.
5	energ.	acous.	speech.	speech.	speech.
6	speech.	speech.	energ.	val.	acous.
7	val.	val.	val.	acous.	val.
8	dance.	energ.	acous.	energ.	energ.
9	dur.	dur.	dur.	dur.	dur.
10	tempo	tempo	tempo	tempo	tempo
11	live.	live.	live.	live.	live.

4 Classification: k-Nearest Neighbour (k=5)

4.1 Implementation

In this second phase, we evaluate the dimensionality reduction techniques previously discussed—PCA, KPCA, and Information Feature Selection—by integrating them into a classification model. Our goal is to determine which method best prepares the data for predicting the music_genre.

For this purpose, we implemented a k-Nearest Neighbor ($k = 5$) classifier. To ensure a fair comparison, we restricted the input to four dimensions for each method,[1] including a reference trial projected onto four arbitrary original variables: danceability, energy, acousticness, and instrumentalness.

The test data is projected into the reduced feature space defined by each method, and the five closest neighbors from the training set are identified using a kd-tree algorithm[2]. Classification is determined by a majority vote; in the event of a tie, neighbors are weighted by their distance to the test point to determine the final genre [3].

4.2 Performance through Subsets

The classification results for each method are summarized in Table 3. The highest performance was achieved using the full set of original features. This outcome is expected, as this approach retains all available information; however, it serves primarily as a performance "ceiling" rather than a scalable solution for larger datasets. Given that this baseline achieved only 50% accuracy, the classifier's overall performance is modest, though it must be noted that this is a 10-class classification problem.

Interestingly, the Information Theoretic Feature Selection methods consistently outperformed the "data compression" approaches (PCA and KPCA). This suggests that for this specific dataset, isolating features with high class-discriminative power is more effective than attempting to capture unsupervised global variance through linear or non-linear transformations.

Overall, the performance of the PC methods remained remarkably similar, appearing largely independent of the specific amount of variance they compressed. The notable exception was the Polynomial KPCA, which performed surprisingly poorly despite capturing nearly all variance in its first component. This suggests a form of "information overfitting": by concentrating the data so heavily into a single axis, the remaining dimensions became redundant, obscuring the boundaries between classes. This outcome highlights a limitation in our optimization strategy; while we maximized cumulative variance for computational efficiency, an ideal—though more resource-intensive—approach would have been to optimize kernel parameters based directly on classifier accuracy.

[1] The baseline run utilizing all original components is the sole exception.

[2] We utilized the built-in kd-tree implementation in R for computational efficiency, as developing a custom spatial indexing algorithm was beyond the scope of this project.

[3] we applied an hybrid approach as normally it's only used one of these comparison measures

Table 3: Knn performance across all strategies. All with only dimension 4, except `all features`. The `4 features` were `danceability,energy, acousticness, instrumentalness`

Method / config	Acc.	Mac. Re	Mac. Pr	Mac. F1
all features	0.4951	0.4952	0.4988	0.4953
4 features	0.2867	0.2864	0.2819	0.2836
PCA	0.3402	0.3395	0.3369	0.3378
RBF KPCA	0.3433	0.3426	0.3389	0.3402
Polynomial KPCA	0.2522	0.2514	0.2481	0.2495
Laplacian KPCA	0.3639	0.3633	0.3618	0.3621
MIM	0.3597	0.3590	0.3624	0.3601
MIFS	0.3295	0.3296	0.3353	0.3310
mRMR, CMIM, JMI	0.4135	0.4133	0.4174	0.4143
maxMIFS	0.3927	0.3923	0.3955	0.3929
DISR	0.3983	0.3979	0.3987	0.3966
DMIM	0.4031	0.4027	0.4025	0.4016

5 Classification: Random Forest

5.1 Training and Evaluation Protocol

A Random Forest (RF) classifier was trained with 500 trees, $m = \lfloor\sqrt{p}\rfloor$ features sampled at each split, and no constraint on minimum leaf size (full-depth trees to minimise bias). Training was performed separately for each dimensionality-reduction strategy: (i) the full original feature set (11 numeric + 2 categorical); (ii) the PCA-reduced representation; (iii) the Kernel PCA-reduced representation and (iv) the subset selected by each of the eight MI methods. All models were evaluated on the held-out test set ($N_{\text{test}} = 15\,000$) using Accuracy, Macro Recall, Macro Precision, and Macro F1, where macro-averaging weights each of the 10 classes equally.

5.2 Results

Table 4 consolidates all strategies and metrics. The full-feature RF leads on every metric. All eight MI FS methods form a tight cluster between Macro F1 0.461 and 0.523, substantially above PCA and Kernel PCA (0.387 and 0.384).

Original dataset. Macro F1 **0.556**; OOB error 0.451 is consistent with the test error, confirming no substantial overfitting. The confusion matrix (Figures 4 and 5) reveals characteristic error clusters: Hip-Hop and Rap are frequently confused (overlapping speechiness and danceability); Blues and Jazz share a cluster (common harmonic texture); Classical is the most reliably identified genre due to its near-binary instrumentalness signature.

Table 4: RF performance across all strategies. FS methods retain 83%–94% of the full-model Macro F1 with 6 features.

Method / config	Acc.	Mac. Re	Mac. Pr	Mac. F1
all features	0.5566	0.5569	0.5593	**0.5558**
CMIM	0.5229	0.5231	0.5258	0.5225
DISR	0.5111	0.5111	0.5138	0.5105
maxMIFS	0.5089	0.5090	0.5110	0.5081
mRMR	0.5084	0.5085	0.5108	0.5072
MIM	0.4909	0.4908	0.4951	0.4912
JMI	0.4883	0.4883	0.4928	0.4888
DMIM	0.4881	0.4881	0.4907	0.4867
MIFS	0.4625	0.4626	0.4644	0.4608
PCA	0.3904	0.3898	0.3872	0.3871
Kernel PCA	0.3868	0.3861	0.3845	0.3837

Feature selection methods. CMIM achieves the best FS performance (Macro F1 0.5225, retaining 93.9% of the full-model score with only 6 features). Its scale-invariant penalisation via maximisation prevents the relevance signal from being overwhelmed. MIFS is the weakest (0.4608), directly validating [**?**]: its unscaled penalty $\sum_s I(X_i; X_s)$ grows linearly with $|S|$ and eventually dominates, selecting maximally independent but less genre-informative features. mRMR and maxMIFS produce near-identical results (0.5072 and 0.5081), consistent with their algorithmic similarity. JMI and DMIM underperform their theoretical standing, likely because conditional MI estimates $I(X_i; X_s \mid C)$ are noisier on fine probability tables required by 10-class discretisation [4]. Confusion matrices for the best and worst FS methods are shown in Figure 4 and 5.

5.3 The Hip-Hop / Rap Separation Problem

The binary feature distribution analysis (Figure 9) reveals why Hip-Hop and Rap account for the largest off-diagonal mass in every confusion matrix. Across all 11 audio features the two genres are essentially indistinguishable The MIM score for the top-ranked feature (`popularity`) is only 0.217, compared to values exceeding 0.4 for the Classical/Rap separation reported in the preliminary analysis.

This suggests that Hip-Hop and Rap are not acoustically distinct genres in this dataset, and so the classifier is correctly reflecting the limits of the available information.

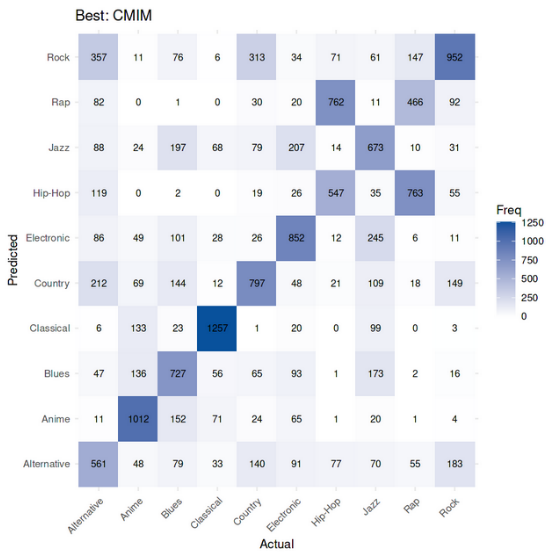

Figure 7: Confusion matrix for CMIM (Macro F1 = 0.522). Hip-Hop/Rap and Blues/Jazz remain the dominant confusion pairs.

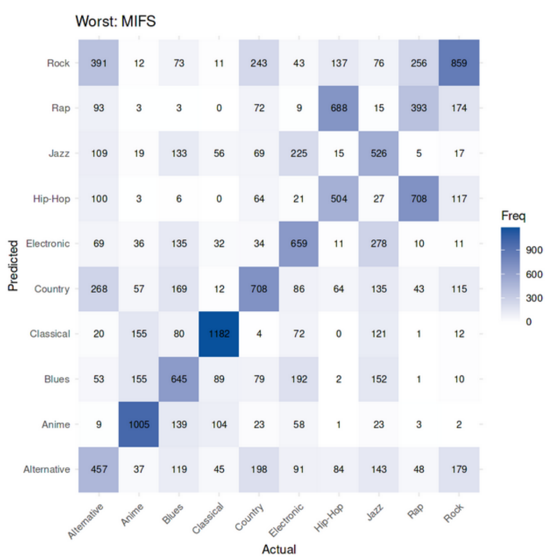

Figure 8: Confusion matrix for MIFS (Macro F1 = 0.461). Increased Hip-Hop/Rap confusion reflects a less discriminative feature subset.

Dimensionality reduction. PCA and Kernel PCA substantially underperform all MI-based approaches.

5.4 Variable Importance vs. MI Rankings

Table 5 compares RF permutation importance rank with MIM (univariate MI) rank. The Spearman rank correlation is 0.691, indicating moderate

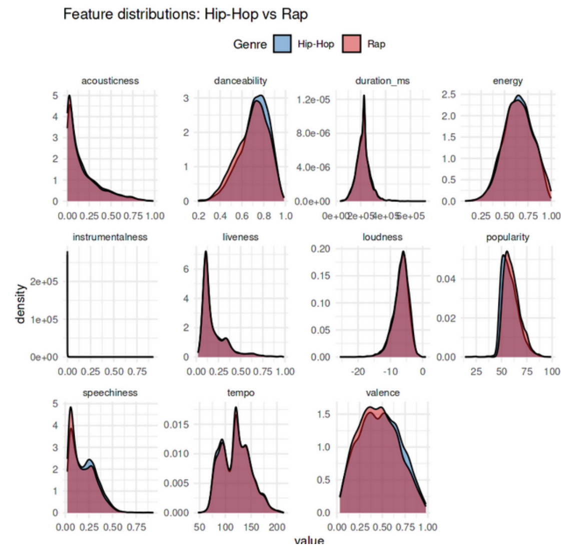

Figure 9: Feature distributions for Hip-Hop vs Rap. Near-complete overlap across all 11 features explains the systematic confusion between these genres in every classifier evaluated.

agreement.

Table 5: RF permutation importance rank vs. MIM rank. Spearman $\rho = 0.691$. Discrepancies ≥ 3 positions are italicised.

Feature	RF Rank	MIM Rank
popularity	1	1
instrumentalness	2	4
danceability	3	*6*
speechiness	4	*7*
loudness	5	*2*
valence	6	*8*
acousticness	7	3
duration_ms	8	9
energy	9	5
tempo	11	10
liveness	12	11

The two most instructive discrepancies are `loudness` (MIM rank 2, RF rank 5) and `danceability` (MIM rank 6, RF rank 3). Loudness carries high marginal MI but is largely redundant once `popularity` and `energy` are in the ensemble—a relationship MIM cannot detect. Danceability becomes more informative in the context of other features than its marginal MI implies, a complementarity effect that CMIM and DMIM are specifically designed to capture.

6 Causal Discovery and Interventional Analysis

6.1 Motivation

Standard classification evaluation does not answer wether the learned feature–label associations are causally stable or spurious correlations that break under distribution shift. Motivated by [5], we complement the benchmarks with three causal tools: a data-driven causal skeleton (PC algorithm), interventional response curves via Pearl's do-calculus, and an Interventional Robustness Score (IRS) proxy.

6.2 Causal Graph via the PC Algorithm

The PC algorithm was applied to the 6 scaled numeric features plus an integer-encoded genre node, using the Gaussian CI test ($\alpha = 0.001$, a conservative value that hardens the maintenance of causal links in the algorithm).

The graph (Figure 7) recovers some interpretable relationships, namely Speechiness and danceability that share an undirected edge, consistent with both being elevated in Hip-Hop and Rap for production-style reasons rather than a direct causal chain. Several edges remain unoriented, displaying purely observational data. An interesting direction for further work would be to analyse the popularity node in more detail and determine which audio characteristics are most strongly associated with higher popularity, both in terms of Spotify's algorithmic ranking and listener perception.

The main changes from the old text: updated the node count (12→7), updated alpha and justified the CMIM restriction, replaced the single vague interpretability sentence with a systematic edge-by-edge reading, and added the genre_node→danceability observation which is the most analytically interesting result in the new graph.

6.3 Interventional Response Curves

For three feature–genre pairs we computed the observational curve $P(\text{Genre} \mid X = t)$ and the interventional curve $P(\text{Genre} \mid \mathbf{do}(X = t))$ as t is swept from the 5th to the 95th percentile. The observational curve is estimated by conditioning on the nearest training point to t; the interventional curve uses the backdoor adjustment:

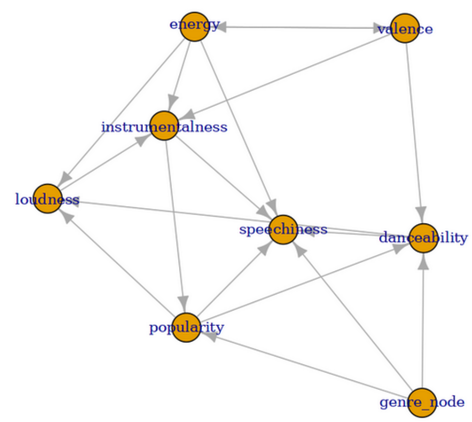

Figure 10: PC-algorithm CPDAG over the six CMIM-selected features and `genre_node` ($\alpha = 0.001$, Gaussian CI test). Oriented edges indicate an identifiable causal direction; unoriented edges are non-identifiable from observational data alone.

$$P(Y \mid \mathbf{do}(X = t)) = \frac{1}{N}\sum_{i=1}^{N} \hat{P}(Y \mid X = t, \mathbf{Z} = \mathbf{z}_i)\,, \tag{1}$$

marginalising over the empirical distribution of remaining features $\mathbf{Z}$ [5]. The RF serves as $\hat{P}$. The single-observation conditioning makes the observational curves jagged; the interventional curves are smooth by construction, and this contrast itself reveals the confounding structure.

Experiment A — Instrumentalness → Classical. Both curves rise with instrumentalness, but the observational curve is markedly more erratic, indicating confounding from co-occurring Classical markers (low loudness, low tempo variability). The backdoor-adjusted curve isolates the direct causal effect.

Experiment B — Speechiness → Rap. The interventional curve shows a smooth monotone increase in predicted Rap probability, confirming a genuine causal contribution after confounders are removed. The smaller gap between the two curves relative to Experiment A suggests speechiness is a more isolated driver of Rap, with fewer co-occurring confounders.

Experiment C — Popularity → Classical. The interventional curve is nearly flat (≈ 0.10–0.15 across the full range), the key diagnostic result: popularity does not causally determine genre. The observational correlation exists because Classical music has characteristically low streaming popularity, but forcing a Classical track to high popularity would not change the classifier's prediction.

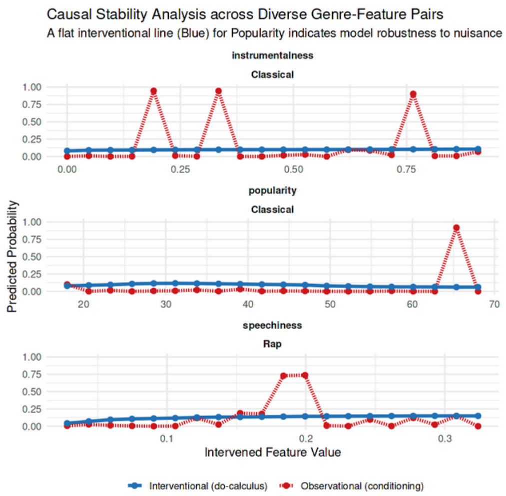

Figure 11: Observational vs. interventional probability curves for three feature–genre pairs. Jagged red curves reflect single-point conditioning; smooth blue curves are the backdoor-adjusted causal estimates (Eq. 1).

6.4 Interventional Robustness Scores

For method m, the IRS proxy is:

$$\widehat{\text{IRS}}_m = 1 - \frac{\overline{\text{PIDA}}_m}{\text{sd}[\hat{P}(\text{Rap} \mid X)]}, \tag{2}$$

where $\overline{\text{PIDA}}_m$ is the mean, over nuisance features in the selected subset, of the maximum absolute shift in predicted Rap probability when that feature is swept through a 5-point quantile grid while the top-ranked feature is held at its median.

Table 6 reports IRS and Macro F1 for all eight methods, revealing an *inverse* relationship between the two quantities. THe IRS values show the model is quite robust across all methods.

7 Conclusions

The full-feature RF (Macro F1 0.556) outperforms all FS models, so it is not harmed by weak features. The value of FS here lies in interpretability, computational efficiency, and causal auditing. To assess sensitivity to training set size, the experiment was repeated with a 90/10 split. All Macro F1 scores decreased slightly (e.g. the full-model RF dropped from 0.556 to 0.544), and the ranking of all strategies was preserved exactly, suggesting that the reported estimates carry small variance under different partitions of the dataset.

Table 6: IRS proxy and Macro F1 by FS method (sorted by IRS, descending). All methods exceed IRS 0.96. The inverse relationship with accuracy is a structural consequence of each method's redundancy-control philosophy.

Method	IRS proxy	Macro F1
MIFS	0.978	0.461
DMIM	0.971	0.487
MIM	0.969	0.491
JMI	0.969	0.489
mRMR	0.965	0.507
maxMIFS	0.965	0.508
DISR	0.964	0.511
CMIM	0.964	0.523

The kNN classifier mirrors the RF ordering across reduction strategies: MI-based feature selection consistently outperforms both PCA and Kernel PCA. The Polynomial KPCA result (Macro F1 0.250) is the sharpest illustration of the variance discriminability mismatch: concentrating nearly all variance into a single axis renders remaining dimensions redundant and collapses class boundaries rather than revealing them.

The RF outperforms kNN by approximately 0.10 Macro F1 across every reduction strategy, a gap that is consistent and strategy-independent. Part of this gap is attributable to the curse of dimensionality: in high-dimensional spaces, Euclidean distances concentrate and the notion of a nearest neighbour becomes increasingly unreliable, as all points tend toward the same pairwise distance. This makes kNN particularly sensitive to the input representation

A Appendix: tables and graphs

Table 7: RBF Kernel Parameter Search

Combo	Parameter (γ)	Cum. Var (2-axis)
1	**0.001**	**0.4521**
2	0.010	0.4005
3	0.050	0.2564
4	0.100	0.1744
5	1.000	0.0120
6	10.00	0.0006

Table 8: Polynomial Kernel Parameter Search

Combo	Parameters (d, c)	Cum. Var (2-axis)
1	1, 0.0	0.4572
2	2, 0.0	0.8142
3	3, 0.0	0.9938
4	**4, 0.0**	**0.9998**
5	1, 0.1	0.4572
6	2, 0.1	0.8121
7	3, 0.1	0.9938
8	4, 0.1	0.9998
9	1, 1.0	0.4572
10	2, 1.0	0.7948
11	3, 1.0	0.9932
12	4, 1.0	0.9998
13	1, 5.0	0.4572
14	2, 5.0	0.7348
15	3, 5.0	0.9903
16	4, 5.0	0.9997

Table 9: Laplacian Kernel Parameter Search

Combo	Parameter (γ)	Cum. Var (2-axis)
1	**0.001**	**0.2876**
2	0.010	0.2710
3	0.100	0.1535
4	1.000	0.0033
5	10.00	0.0006

The following table details the cumulative explained variance ratio for standard PCA and the three investigated Kernel PCA (KPCA) variants as the number of principal components (axes) increases.

Table 10: Cumulative Explained Variance by Principal Component Axis

Method—Info% with	1 dim	2 dim	3 dim	4 dim
PCA	0.3364	0.4590	0.5542	0.6433
KPCA (RBF)	0.3354	0.4541	0.5465	0.6333
KPCA (Poly)	0.9986	0.9994	0.9997	0.9998
KPCA (Laplace)	0.2103	0.2887	0.3468	0.3960

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